

Sanskrit Retrieval-Augmented Generation (RAG) System

Overview

This project is an end-to-end Retrieval-Augmented Generation (RAG) system designed for Sanskrit documents.

The primary objective was to build a retrieval pipeline capable of handling Sanskrit queries written in multiple scripts while maintaining high retrieval quality through systematic evaluation and experimentation.

Unlike many RAG projects that focus only on generation, this project places significant emphasis on retrieval quality. A custom benchmark was created, retrieval metrics were implemented from scratch, and multiple retrieval strategies were evaluated through ablation studies.

Motivation

Sanskrit presents several challenges that are uncommon in modern-language retrieval systems:

- Multiple writing schemes (Devanagari, IAST, Harvard-Kyoto, SLP1)
- Compound words and sandhi formations
- Limited availability of Sanskrit-specific retrieval benchmarks
- Cross-script query/document matching

The goal was to build a system that could:

- Accept Sanskrit queries in different transliteration schemes
- Retrieve relevant passages reliably
- Generate grounded answers from retrieved context
- Measure retrieval quality using standard IR metrics

Corpus

The initial corpus consists of Sanskrit narrative passages and educational stories including:

- मूर्खभृत्यस्य
- चतुरस्य कालिदासस्य
- वृद्धायाः चातुर्यम्
- देवभक्तः
- शीतं बहु बाधति

Documents are automatically cleaned, normalized, segmented into verses, and indexed for retrieval.

System Architecture

Query ↓ Script Detection ↓ Transliteration Normalization ↓ Hybrid Retrieval

├── FAISS Semantic Search └── BM25 Keyword Search

↓ Reciprocal Rank Fusion (RRF) ↓ CrossEncoder Re-ranking ↓ Context Selection ↓ Llama 3.2 Local Inference ↓ Answer Generation

Retrieval Pipeline

Transliteration-Aware Retrieval

Queries may be entered in:

- Devanagari
- IAST
- Harvard-Kyoto (HK)
- SLP1

All queries are automatically normalized to Devanagari before retrieval.

Example:

vRddhA vanam gatvA kim apazyat?

↓

वृद्धा वनं गत्वा किम् अपश्यत्?

This significantly improves retrieval consistency across scripts.

Hybrid Retrieval

Two retrieval methods operate in parallel:

Dense Retrieval

Model:

intfloat/multilingual-e5-small

Vector Store:

FAISS IndexFlatIP

Sparse Retrieval

Algorithm:

BM25Okapi

The combination allows both semantic similarity and exact phrase matching.

Reciprocal Rank Fusion

Dense and sparse rankings are merged using Reciprocal Rank Fusion (RRF).

This proved more robust than relying on either retrieval method individually.

Neural Re-ranking

The top fused candidates are re-ranked using:

cross-encoder/mmarco-mMiniLMv2-L12-H384-v1

This stage improved ranking quality substantially and produced strong MRR scores.

Evaluation Methodology

A custom benchmark containing 21 manually annotated Sanskrit queries was created.

Each benchmark entry contains:

- Query
- Relevant Verse Range(s)
- Expected Answer

Retrieval quality is evaluated using:

- Precision@5
- Recall@5
- Mean Reciprocal Rank (MRR)

- NDCG@5
- Mean Average Precision (MAP)

Evaluation is performed at the verse level rather than raw chunk level.

Experiments and Findings

Several retrieval configurations were tested.

Chunking Experiments

Configuration	Precision@5	Recall@5	MRR	NDCG@5	MAP
4 Verse + 1 Overlap	0.4667	0.6667	0.9365	0.7105	0.5973
3 Verse + 0 Overlap	0.2857	0.5556	0.9206	0.6167	0.5126
2 Verse + 0 Overlap	0.3333	0.4794	0.8730	0.5390	0.4031

Key Observation

Contrary to expectations, smaller chunks reduced overall retrieval quality.

The best-performing configuration used:

- 4 verses per chunk
- 1 verse overlap

This provided stronger contextual coherence and consistently higher ranking metrics.

Challenges Encountered

Transliteration Retrieval Failures

Early evaluation revealed complete retrieval failures for certain transliterated Sanskrit queries.

Investigation showed inconsistencies between Romanized query forms and indexed Devanagari text.

To address this:

- Script detection was introduced
- Query normalization was improved
- Unicode normalization was applied consistently

This significantly improved retrieval robustness.

Final Results

Evaluation Set:

21 manually annotated Sanskrit queries

Results:

Metric	Score
Precision@5	0.4667
Recall@5	0.6667
MRR	0.9365
NDCG@5	0.7105
MAP	0.5973

The high MRR score indicates that relevant passages are typically retrieved at the top of the ranking.

Future Work

Potential improvements include:

- BGE-M3 embeddings
 - multilingual-e5-large embeddings
 - multilingual rerankers
 - Sanskrit-specific embedding fine-tuning
 - Query expansion and HyDE retrieval
 - Larger benchmark datasets
 - Larger Sanskrit corpora
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Key Takeaway

The most valuable outcome of this project was not simply building a Sanskrit RAG pipeline, but systematically evaluating and improving it through experimentation.

The final architecture was selected through retrieval benchmarking, error analysis, and ablation studies rather than intuition alone.